

practice_test2

 This is a preview of the published version of the quiz

Started: Apr 7 at 12:25pm

Quiz Instructions

Its open book. does not count toward final grade



Question 1 8 pts

The electric field at a point 2.8 cm from a small object points toward the object and is equal to 180,000 N/C. What is the object's charge q ? ($k = 1/4\pi\epsilon_0 = 8.99 \times 10^9 \text{ N} \cdot \text{m}^2/\text{C}^2$)

hint: The electric field is toward the charge, so the charge is negative.

remember to convert cm to m. Use the inverse square law. $E = K q / (r^2)$

☐

+17 nC

☐

-17 nC

☐

-16 nC

☐

+16 nC



Question 2 8 pts

Two charges separated by one meter exert 1-N forces on each other. If the charges are pulled 3 meters apart, the force on each charge will be

hint : this is inverse square law. you can give yourself $q_1=q_2=1\text{C}$ $d=1\text{m}$ then $d = 3\text{m}$ and $k=1$

☐

3 N.

☐

0.33 N.

☐

9 N.

☐

0 N.

☐

0.11 N.



Question 3 8 pts

Two charged particles repel each other with a force F . If the charge of one of the particles is doubled and the distance between them is halved, then the force will be

hint: you can give yourself $q_1=q_2=1\text{C}$ and $d=1\text{m}$ then $q_1=1\text{C}$ $q_2=2\text{C}$ and $d=0.5$. $k=1$

☐

$F/2$.

☐

F .

☐

none of the above

☐

$F/4$.

☐

$2F$.



Question 4 8 pts

A proton is placed in an electric field $E = 800 \text{ N/C}$. What are the magnitude and direction of the acceleration of the proton due to this field? ($e = 1.60 \times 10^{-19} \text{ C}$, $m_{\text{proton}} = 1.67 \times 10^{-27} \text{ kg}$)

hint: use Newton's second law. $F = ma$ and $F = qE$.

A proton moves with the field. From high potential to low potential. An electron does the opposite.

☐

$76.6 \times 10^{10} \text{ m/s}^2$ opposite to the direction of the  field

☐

$7.66 \times 10^{10} \text{ m/s}^2$ in the direction of the  field

☐

$76.6 \times 10^{10} \text{ m/s}^2$ in the direction of the  field

☐

$7.66 \times 10^9 \text{ m/s}^2$ opposite to the direction of the  field

☐

$7.66 \times 10^{10} \text{ m/s}^2$ opposite to the direction of the  field



Question 5 8 pts

A $+5.0\text{-}\mu\text{C}$ point charge is placed at the 0 cm mark of a meter stick and a $-4.0\text{-}\mu\text{C}$ charge is placed at the 50 cm mark. What is the net electric field field at the 30 cm mark? ($k = 1/4\pi\epsilon_0 = 8.99 \times 10^9 \text{ N} \cdot \text{m}^2/\text{C}^2$)

hint: Make a drawing and use the superposition principle. Place your charges on an axis. The electric

field at point P (30cm mark) is the Sum of E1 away from the +5uC and of E2 toward the -4uC. So MAKE sure you find the net vector (make a drawing. If they points to the same direction, add them. Otherwise, subtract).

☐

$$4.0 \times 10^5 \text{ N/C}$$

☐

$$9.0 \times 10^5 \text{ N/C}$$

☐

$$5.0 \times 10^5 \text{ N/C}$$

☐

$$1.4 \times 10^6 \text{ N/C}$$

☐

Question 6 8 pts

Two tiny particles carrying like charges of the same magnitude are 1.0 mm apart. If the electric force exerted on one of them is 5.0 N, what is the magnitude of the charge on each of these particles? ($k = 1/4\pi\epsilon_0 = 9.0 \times 10^9 \text{ N} \cdot \text{m}^2/\text{C}^2$)

hint: use coulombs law (force is the same). $K q_1 q_2 / d^2$

☐

$$2.4 \times 10^{-5} \text{ C}$$

☐

$$4.5 \times 10^4 \text{ C}$$

☐

$$2.4 \times 10^{-2} \text{ C}$$

☐

$$2.4 \times 10^{-8} \text{ C}$$

☐

Question 7 8 pts

An ideal air-filled parallel plate capacitor with plate a separation of 4.0 cm has a plate area of 0.040 m^2 . What is the capacitance of this capacitor with air between these plates? ($\epsilon_0 = 8.85 \times 10^{-12} \text{ C}^2/\text{N} \cdot \text{m}^2$)

hint: use the equation. See the equation sheet.

☐

$$8.9 \mu\text{F}$$

☐

$$8.9 \text{ pF}$$

☐

$$8.9 \text{ nF}$$

☐

$$89 \text{ pF}$$



0.89 pF



Question 8 4 pts

Two tiny particles having charges $+20.0 \mu\text{C}$ and $-8.00 \mu\text{C}$ are separated by a distance of 20.0 cm. What are the magnitude and direction of the electric field midway between these two charges? ($k =$

$$1/4\pi\epsilon_0 = 9.0 \times 10^9 \text{ N} \cdot \text{m}^2/\text{C}^2)$$

hint: see previous problem. You are adding 2 forces (vectors). See previous comments.


 $25.2 \times 10^6 \text{ N/C}$ directed towards the negative charge

 $25.2 \times 10^5 \text{ N/C}$ directed towards the positive charge

 $25.2 \times 10^4 \text{ N/C}$ directed towards the negative charge

 $25.2 \times 10^6 \text{ N/C}$ directed towards the positive charge

 $25.2 \times 10^5 \text{ N/C}$ directed towards the negative charge


Question 9 8 pts

A $6.0 \mu\text{C}$ negative charge is attracted to a large, well-anchored, positive charge. How much kinetic energy does the negatively charged object gain if the potential difference through which it moves is $\square$

hint: PE = change in KE



0.50 kJ

6.0 μJ 

0.50 J



18 nJ



Question 10 8 pts

What charge accumulates on the plates of a $2.0\text{-}\mu\text{F}$ air-filled capacitor when it is charged until the potential difference across its plates is 100 V?

50 μC 200 μC

100 μC 150 μC **Question 11 8 pts**

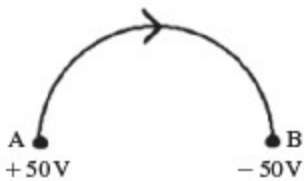
When the magnitude of the charge on each plate of an air-filled capacitor is $4 \mu\text{C}$, the potential difference between the plates is 80 V. What is the capacitance of this capacitor?

20 μF 

50 nF

0.1 μF 100 μF 50 μF **Question 12 8 pts**

How much work must we do on an electron to move it from point A, which is at a potential of +50V, to point B, which is at a potential of -50 V, along the semicircular path shown in the figure? Assume the system is isolated from outside forces. ($e = 1.60 \times 10^{-19} \text{ C}$)



This cannot be determined because we do not know the distance traveled.



1.6 J

 $1.60 \times 10^{-17} \text{ J}$ 

-1.6 J

 $-1.60 \times 10^{-17} \text{ J}$ **Question 13 8 pts**

Each plate of a parallel-plate air capacitor has an area of 0.0040 m^2 , and the separation of the plates is 1.0 mm . An electric field of $1.0 \times 10^6 \text{ V/m}$ is present between the plates. The potential difference across the capacitor is closest to:

☐

467 V

☐

350 V

☐

233 V

☐

583 V

☐

700 V



Question 14 1 pts

MCAT special easy:

To increase the capacitance of a given capacitor, a dielectric is inserted between the two plates. Dielectrics are materials that block the flow of charge. Which of the following materials would be the best dielectric?

☐ **A.** Salt water☐ **B.** Steel, an alloy primarily composed of iron☐ **C.** Glass, an oxide of silica☐ **D.** Copper☐

B

☐

A

☐

C

☐

D



Question 15 1 pts

MCAT easy

-

Two parallel conducting plates are separated by a distance d . One plate carries a charge $+Q$ and the other carries a charge $-Q$. The voltage between the plates is 12 V. If a $+2 \mu\text{C}$ charge is released from rest at the positive plate, how much kinetic energy does it have when it reaches the negative plate?

☐ A. $2.4 \times 10^{-6} \text{ J}$ ☐ B. $4.8 \times 10^{-6} \text{ J}$ ☐ C. $2.4 \times 10^{-5} \text{ J}$ ☐ D. $4.8 \times 10^{-5} \text{ J}$

Easy. Conservation of energy

☐

A

☐

C

☐

B

☐

D

☒

Question 16 1 pts

MCAT

Question 1: Cell Membrane as a Capacitor

A cell membrane is modeled as a parallel-plate capacitor with a dielectric constant (κ) of 5, a thickness (d) of 8 nm, and an area (A) of $1 \times 10^{-10} \text{ m}^2$. Calculate the capacitance of the membrane. If a disease causes the membrane to thicken to 16 nm, how does the capacitance change?

For the second part of the question . What happens to the capacitance if the distance gets larger? you can play with the simulation: https://phet.colorado.edu/sims/html/capacitor-lab-basics/latest/capacitor-lab-basics_en.html or you can use the equation for capacitance. Make sure to convert properly. Call C the new capacitance and C0 the old one.

☐

5.53 pF and $C = 2C_0$

☐

5.53 uF and $C = C_0/2$

☐

5.53 nF and $C = C_0/4$

☐

5.53 pF and $C = C_0/2$



Question 17 1 pts

BIO - MCAT

Three capacitors (3 μF , 7 μF , and 10 μF) are wired in parallel in a circuit mimicking synaptic vesicle fusion. What is the total capacitance?

hint: I know we didn't do that in class. but it is easy. OPEN your book or do research/ How to find the equivalent capacitance of 3 capacitors in parallel?

Synaptic vesicles release neurotransmitters via calcium-triggered exocytosis. Parallel capacitors model the combined charge storage capacity of vesicles.

☐

20nF

☐

2 uF

☐

20uF

☐

5 uF



Question 18 1 pts

A defibrillator uses a capacitor charged to 2000 V. If the capacitor has a capacitance of 50 μF , calculate the energy stored. How does adding a dielectric ($\kappa = 2$) affect the energy?

Can the old energy U_0 and the new U ?

☐

$U_0 = 100\text{J}$, $U = U_0/2$

☐

$U_0 = 100\text{J}$. $U = U_0$

☐

$U_0 = 100\text{J}$. $U = 2U_0$

☐

$U_0 = 1000\text{J}$. $U = U_0$

Not saved

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